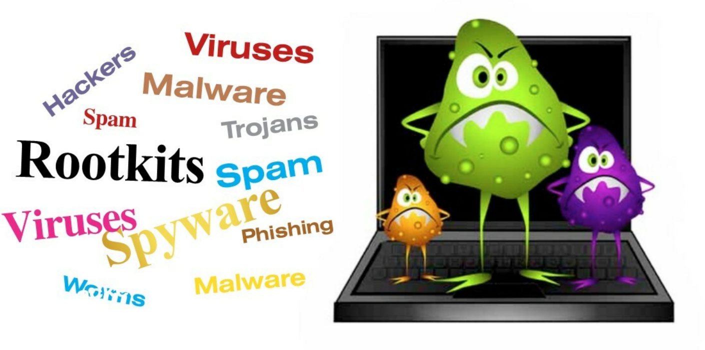

Chkrootkit

Eliminate the Enemy Within

Last month, we learnt about the technical components that form root-kits. We also discussed why and how root-kits are dangerous, and briefly explored a few detection options. This article continues on the same topic, covering proven methods to detect the presence of root-kits on servers—a powerful open source utility named Chkrootkit, dedicated to this purpose. We cover its installation and usage, and ways to incorporate it in daily administrative tasks, to ensure the security of servers and data centres.

In the past several years, open source Linux distributions have emerged as a rock solid platform for production data centres with mission-critical IT infrastructure. This growth has prompted hackers to try exploits to 'own' Linux systems; the most serious being root-kits, which are far more dangerous than viruses and Trojans. Since we covered the basics in the previous article in this series, please check the September 2011 issue of *LFY* for that information.

About Chkrootkit

Chkrootkit is a collection of tools to detect the presence of root-kits, and is a gift to Linux systems administrators for two specific reasons: first, it is a free, open source utility, and available for multiple distros. Second, it detects almost all the latest root-kits out there, since the open source contributors' community keeps it up to date. Over time, the *Chkrootkit* scan engine has also improved, making it faster, which is especially useful in performing detailed kernel checks against a number of supported kit detections.

A few great features of *Chkrootkit* are that it detects more than 60 old and new kits, is capable of detecting network interfaces in promiscuous mode, can efficiently detect altered *lastlog* and *wtmp* files (which in turn alerts admins about intrusions), has easy command-line access with straightforward options, and has a verbose output mode to help admins automate tasks.

Chkrootkit uses C and shell scripts to perform a detailed process check, and scans systems binaries to detect kit signatures. Upon detection, in most cases, it can remove root-kits too. *Chkrootkit* also has a few algorithms that can report trends of a possible root-kit, even if it is not yet officially supported. Table 1 lists the *Chkrootkit* internal programs and what each of them do.